MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 9 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. Let $f(x)=\int_0^x \sin\left(t^2-t+x\right)dt$, then value of $f''\left(-\frac{1}{2}\right)+f\left(-\frac{1}{2}\right)$ is

(a) 0 (b) π (c) $\frac{\pi}{2}$ (d) $\frac{3\pi}{2}$

2. Lines $(1+\lambda)x+(4-\lambda)y+(2+\lambda)=0$ and $(4-\lambda)x-(1+\lambda)y+(6-3\lambda)=0$ are concurrent at points *A* and *B* respectively and intersect at *C*, then locus of centroid of ΔABC is

(a) $\left(x+\frac{3}{2}\right)^2+\left(y+\frac{7}{10}\right)^2=\frac{34}{225}$ (b) $\left(x-\frac{3}{2}\right)^2+\left(y-\frac{7}{10}\right)^2=\frac{34}{225}$

(c) $\left(x-\frac{3}{2}\right)^2+\left(y+\frac{7}{10}\right)^2=\frac{34}{225}$ (d) none of these

3. $f:[-1,1]\to R$ such that $f(0)=f(1)$ and $\forall\, a,b\in[-1,1]\ :\ |f(b)-f(a)|<|b-a|$, then which of the following is always true?

(a) $\exists c\in(-1,1): f'(c)>1$ (b) $|f(b)-f(a)|<1/2$

(c) $f(x)\le x$ (d) $f(b)\ne f(a)\ \forall a\ne b$.

4. If 84^n divides $(496)!$, then the maximum value of *n* can be

(a) 160 (b) 40 (c) 80 (d) none of these

5. The fundamental period of function $f(x)=\cos^7 x+\cos^7\left(x+\frac{2\pi}{3}\right)+\cos^7\left(x+\frac{4\pi}{3}\right)$ is

(a) 2π (b) $\frac{2\pi}{3}$ (c) $\frac{4\pi}{3}$ (d) $\frac{\pi}{3}$

6. One side of square *ABCD* is on the line $y = 2x - 17$ and other two vertices are on parabola $y = x^2$, then the minimum area of square is

(a) 80 (b) $4\sqrt{5}$ (c) 1280 (d) $16\sqrt{5}$

7. The length of the perpendicular from the origin to the plane passing through three non-collinear points $\vec{a}, \vec{b}, \vec{c}$ is

(a) $\dfrac{[\vec{a}\,\vec{b}\,\vec{c}]}{|\vec{a}\times\vec{b}+\vec{c}\times\vec{a}+\vec{b}\times\vec{c}|}$ (b) $\dfrac{2[\vec{a}\,\vec{b}\,\vec{c}]}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}$

(c) $[\vec{a}\,\vec{b}\,\vec{c}]$ (d) none of these

8. Let O be an interior point of ΔABC such that $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$, then the ratio of area of ΔABC to area of ΔAOC is

(a) 2 (b) $\dfrac{3}{2}$ (c) 3 (d) $\dfrac{5}{3}$

9. The number of 5 digit multiples of 3 that end with digit 6 is

(a) 30000 (b) 6000 (c) 2999 (d) 3000

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

10. A drawer contains red and black balls. When two balls are drawn at random, the probability that they both are red is $\frac{1}{2}$. The number of balls in the drawer can be

(a) 21 (b) 11 (c) 4 (d) 3

11. If a function satisfies $(x-y)f(x+y)-(x+y)f(x-y)=2(x^2y-y^3)\ \forall\ x, y \in R$ and $f(1)=2$, then

(a) $f(x)$ must be polynomial function (b) $f(3)=12$

(c) $f(0)=0$ (d) $f(x)$ may not be differentiable

12. $f:[1,4]\to[7,14]$ is a surjective, twice differentiable function such that $f''(x)>0$, $\forall x\in(1,\ 4)$, then the equation $(f'(x))^2=\frac{49}{9}$ has

(a) at least one root in (1, 4) (b) exactly one root in (1, 4)
(c) at least two roots in (1, 4) (d) at most two roots in (1, 4)

13. In a ΔABC medians AD and BE are perpendicular to each other, then

(a) $\frac{1}{2}<\frac{a}{b}<2$ (b) $\frac{1}{2}<\frac{c}{a}<2$

(c) $\frac{1}{2}<\frac{a}{c}<2$ (d) $\frac{1}{2}<\frac{b}{a}<2$

SECTION – III
Reasoning Type

This section contains 3 reasoning type questions numbered 14 to 16. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.
(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.
(C) **Statement-1** is True, **Statement-2** is False.
(D) **Statement-1** is False, **Statement-2** is True.

14. **Statement-1:** If two points lie on opposite sides of all lines forming a triangle then one of these points has to lie inside the triangle.
Statement-2: If two points lie on opposite side of two intersecting lines then they are in vertically opposite regions.

(a) A (b) B (c) C (d) D

15. **Statement-1 :** If $A=[a_{ij}]_{n\times n}$ and $a_{ij}=i^2-j^2$ then A is necessarily a skew symmetric matrix $n>1$.

Statement-2 : A skew-symmetric matrix of odd order is always invertible.

(a) A (b) B (c) C (d) D

16. **Statement 1 :** If $\sin^{-1}\left(1-y^2\right)+\sin^{-1}\left(1-x\right)=\cos^{-1}\left(1+x^2+y^2+2xy^2\right)$, then locus of $(x,\ y)$ is parabola.

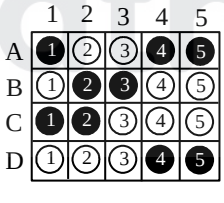
Statement 2 : $\sqrt{1-x-2y}+\sqrt{1-x+2y}=2$ represents a parabola.

(a) A (b) B (c) C (d) D

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1. The tangent to $y^2 = 4ax$ at $P(at^2, 2at)$ intersects axis at T and the normal at P intersects at N, S is the focus and M is foot of perpendicular from P on directrix.

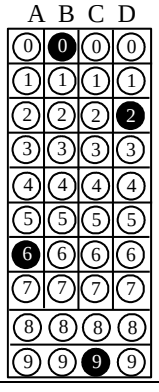
Column -I	Column -II
(A) $a(1+t^2)$	**1.** intercept on normal made by circle with PS as diameter.
(B) $a\sqrt{1+t^2}$	**2.** distance of M from tangent at P.
(C) $\frac{1}{t}$	**3.** distance PN.
(D) $\frac{a}{2}\left(1+t^2\right)$	**4.** $\tan(\angle MPT)$
	5. slope of tangent at P.

2. Let (x, y) be such that $\sin^{-1}(ax)+\cos^{-1}(y)+\cos^{-1}(bxy)=\frac{\pi}{2}$.

Column I	Column II
(A) If $a = 1$ and $b = 0$, then (x, y)	**1.** lies on the circle $x^2+y^2=1$
(B) If $a = 1$ and $b = 1$, then (x, y)	**2.** lies on $\left(x^2-1\right)\left(y^2-1\right)=0$
(C) If $a = 1$ and $b = 2$, then (x, y)	**3.** lies on $4x^2+y^2=1$
(D) If $a = 2$ and $b = 0$, then (x, y)	**4.** lies on the curve having bounds area $\frac{\pi}{2}$
	5. lies on the curve having bounds area π

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



---

1. Let $\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = \frac{1}{6}(x-\alpha_1)(x-\alpha_2)(x-\alpha_3)(x-\alpha_4)$ be an identity in x, where $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ are independent of x. Then find the value of $\alpha_1\alpha_2\alpha_3\alpha_4$.

Space for work

mathongo

---

2. $f:R\to R$ be a twice differentiable function satisfying $f''(x)-5f'(x)+6f(x)\geq 0\ \forall\ x\geq 0$ if $f(0)=1\quad f'(0)=0$. If $f(x)$ satisfies $f(x)\geq ah(bx)-bh(ax),\ \forall x\geq 0$, then find $(a+b)h(0)$.

Space for work

mathongo

3. A sequence is obtained by deleting all perfect squares from set of natural numbers. Find the remainder when the 2003rd term of new sequence is divided by 2048.

Space for work

mathongo

4. Find the least integral value of a such that $\sqrt{9-a^2+2ax-x^2} > \sqrt{16-x^2}$ for at least one positive x.

Space for work

mathongo